

Failure mode measured on the VCS	VCS measured evidence proven in this benchmark	Neural-network analogue documented, not proven here	Status here → there
1 Gradient vanishes on discrete / index outputs	0.234 gradient bucket F (position) single record The NAIVE position gradient is provably zero at a strobe-timed argmax (§3.2); the whole gradient bucket stays low, 0.234, on the position regime.	Shattered / saturated gradients; non-differentiable argmax & index readouts.	proven here analogue there
2 Saliency is model-invariant (sanity-check fails)	0.000 sanity-check margin single record Program-randomization: saliency cosine = 1.000 while 96% of RAM changed -> sanity check FAILS.	Saliency unchanged under model- & label-randomization; behaves like an edge detector.	proven here analogue there
3 Correlation ≠ causation (Granger & tuning trap)	0.136 Granger F 95% CI [0.09, 0.18] Granger-edge faithfulness = 0.136 (false edges vs the true data-flow); game-variable tuning = 0.213.	Spurious Granger edges; 'present ≠ used' tuning & attention read as explanation.	proven here analogue there
4 Single-unit ablation has an interaction blind-spot	0.207 connectome F 95% CI [0.13, 0.29] Single-unit lesion is faithful (0.965) yet misses the wiring: connectome recovered at only 0.207.	Single-unit ablation misleading; computation is distributed across many units.	proven here analogue there
5 SAE/probe: present-not-used	0.108 selectivity F 95% CI [0.08, 0.14] SAEs match a variable, yet score 0.173 on causal use; probes decode at 0.951 acc but 14 cells are not causal.	Control-task selectivity gap; superposition / polysemantic neurons.	proven here analogue there
6 IG/EG baseline sensitivity	0.175 EG F (all-regime) 95% CI [0.11, 0.24] IG magnitude is baseline-sensitive (up to 0.999 across games); a content-byte baseline kills IG; EG lands at 0.175.	Attribution depends on the (arbitrary) baseline choice; no canonical reference input.	proven here analogue there